MINI FUNCTION GENERATOR

Arduino-Based Mini Function Generator

STUDENT DETAILS

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Class : BSc (EMCS)

Semester : 3rd

Subject : Electronics

PROJECT OVERVIEW

I built this mini function generator to produce the common test waveforms we use in electronics labs. A commercial function generator is expensive, so I wanted a simple, low-cost one that I could build myself and use for testing circuits and running experiments.

OBJECTIVE

My goal was to build a compact, low-cost function generator that can produce sine, square, triangle and sawtooth waves, with adjustable frequency and amplitude, so I could use it to test and study other circuits.

MAIN FEATURES

Four waveforms you can switch between with push buttons. Adjustable frequency and amplitude. Arduino-controlled, with a 6-bit R-2R digital-to-analogue converter and LM358 buffering. Low cost, compact and simple to use.

KEY SPECIFICATIONS

Parameter	Details
Waveforms	Sine, square, triangle, sawtooth (selected with push buttons)
DAC	6-bit R-2R resistor ladder, 64 voltage levels
Samples per cycle	64
Controls	Frequency potentiometer, amplitude potentiometer, 4 waveform buttons
Output stage	LM358 unity-gain buffers (U1:A and U1:B)
Power	9 V supply
Output viewed on	CRO / oscilloscope

COMPONENTS USED

Component	Value	Qty
Arduino Uno	Microcontroller	1
LM358	Dual op-amp (U1:A, U1:B)	1
Resistors	10 kΩ (ladder arms, R1, R2)	7
Resistors	22 kΩ (2R legs and termination)	7
Potentiometers	10 kΩ (RV1 frequency, RV2 amplitude)	2
Capacitor	100 nF (C1)	1
Push buttons	Waveform selection	4
Supply and CRO	9 V supply, oscilloscope	1 each

I used 22 kΩ for the 2R value because 20 kΩ resistors were hard to find.

PIN CONNECTIONS

Arduino pin	Connected to
D7 to D2	R-2R ladder, D7 = MSB, D2 = LSB
A0	RV1 wiper (frequency)
A1 / A2 / A3 / A4	Sine / Square / Sawtooth / Triangle buttons (other side to GND)
5 V and GND	Ends of RV1

CONSTRUCTION

Arduino pins D2 to D7 drive the R-2R ladder, with D7 as the most significant bit. The ladder output goes into U1:A, an LM358 buffer. RV1 sets the frequency and is read on A0, and the four buttons (sine, square, sawtooth, triangle) are on A1 to A4. From the buffer, the signal goes through the amplitude pot RV2 to U1:B, a second buffer that drives the CRO.

R2 and C1 are connected from the U1:A output (pin 1) to ground, not in series with the signal, so they don't round off the edges of the square wave.

WORKING PRINCIPLE

The R-2R ladder turns the Arduino's digital values into voltages. Each waveform is stored as 64 samples, and the Arduino sends them to the ladder over and over. It reads RV1 to decide how long to wait between samples, and that sets the frequency. RV2 sets the amplitude, and U1:B buffers the output so the CRO sees a stable signal.

Because the DAC is 6-bit (64 voltage levels), the sine wave shows small steps.

EXPECTED OUTPUT

I expected four stable, clean waveforms with the right shapes and sharp transitions, easy to see on the CRO, with frequency and amplitude that I could adjust.

ACTUAL OUTPUT / RESULT

The prototype worked. I got all four waveforms (sine, square, triangle and sawtooth) on the CRO. The demonstration video shows the real traces.

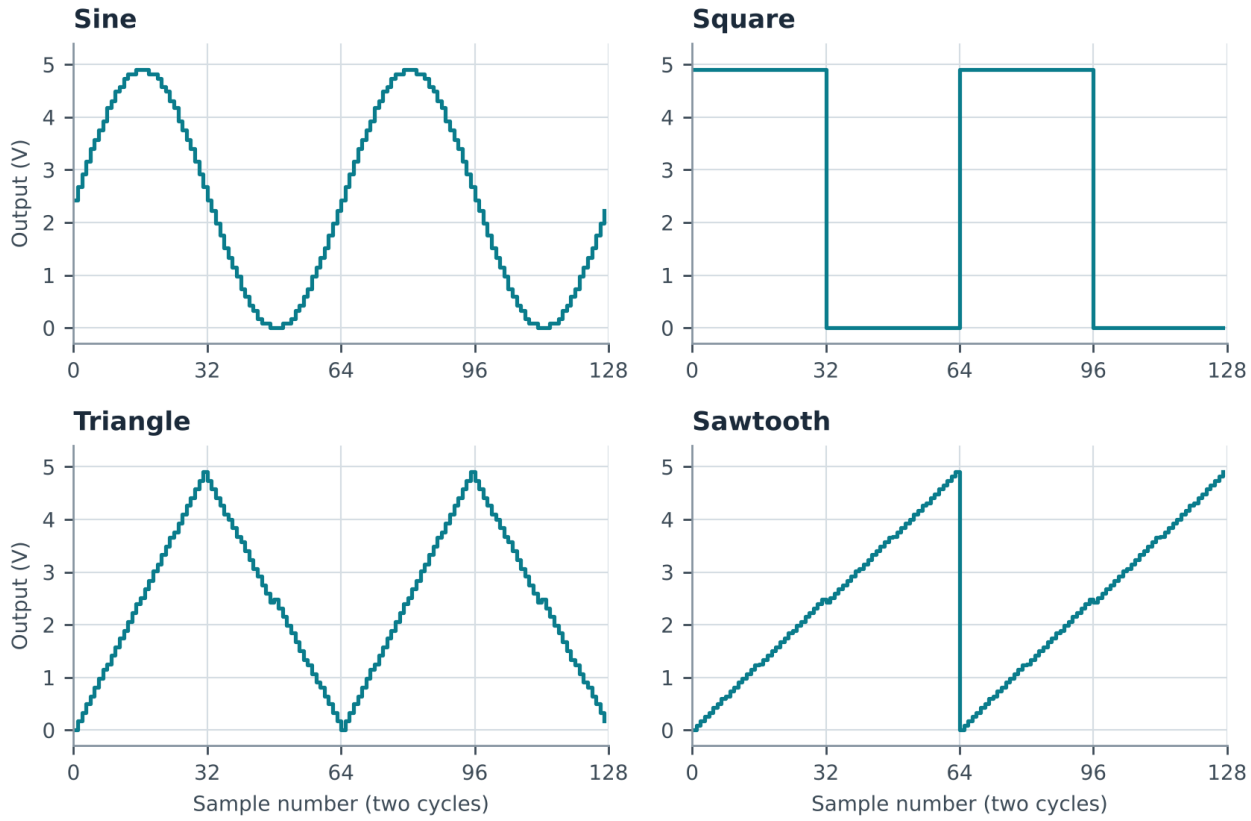
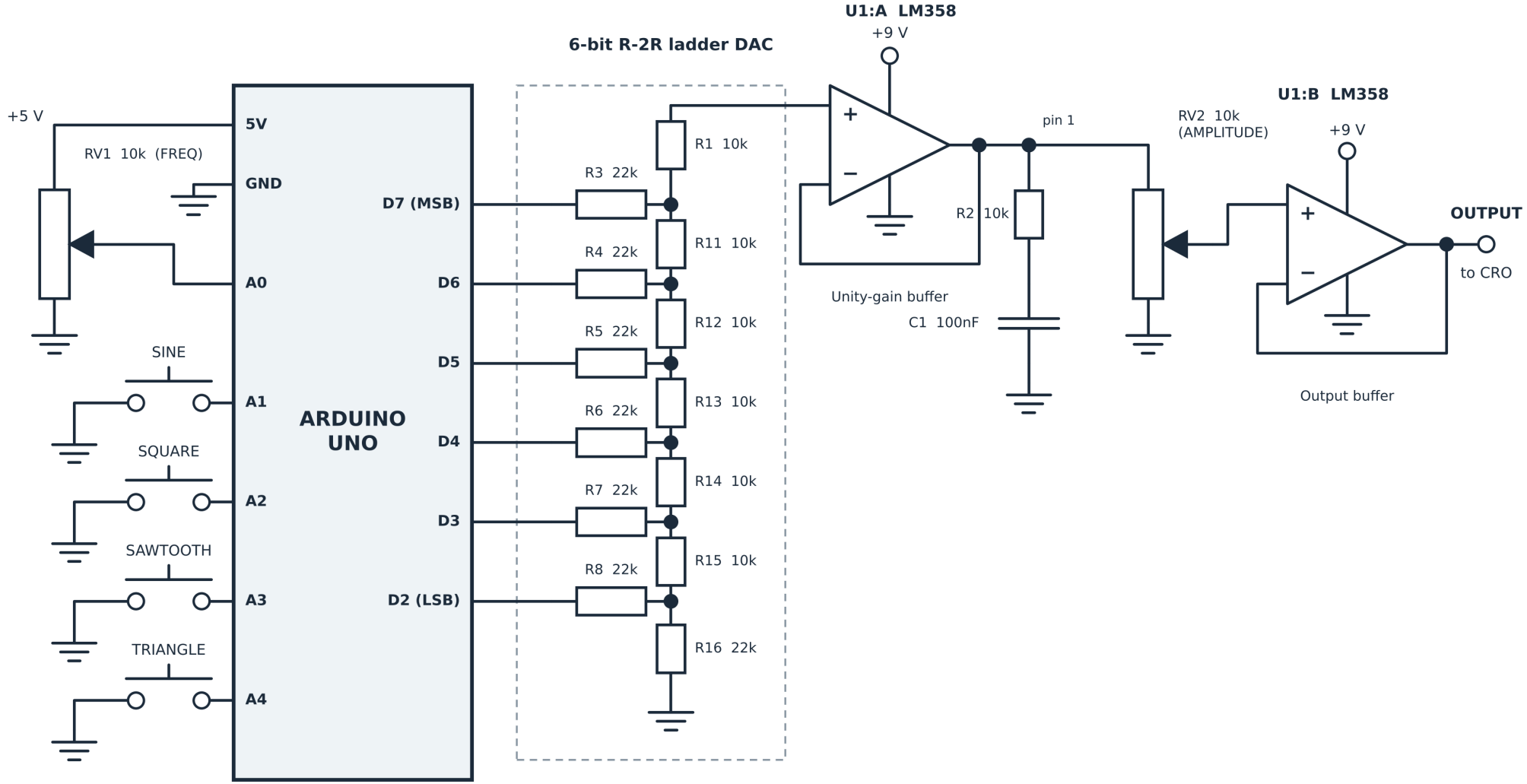


Fig. 1: Simulated DAC output for the four waveforms (64 samples per cycle, 6-bit, 10 kΩ / 22 kΩ ladder), two cycles at full amplitude.

SOURCE CODE

The full Arduino code is on GitHub:
github.com/king2night/Aurduino_Based_Mini_Funtion_Generator

FINAL CIRCUIT DIAGRAM



R = 10 kΩ, 2R = 22 kΩ (standard value used in place of 20 kΩ) | Supply: 9 V (Arduino Vin and LM358 V+) | Buttons: A1 sine, A2 square, A3 sawtooth, A4 triangle (active low to GND)